

More, Fewer, or the Same?

Answer Key

Kindergarten–Grade 1

Quick Answers

- | | |
|---|--|
| 1. cats = 3, bowls = 5, more cats or more bowls (bowls, $3 < 5$) = $<$ | 5. Tam's grapes = 8, Ana's grapes = 6, who has fewer (Ana, $8 > 6$) = $>$ |
| 2. Row A = 6, Row B = 4, which row has fewer (B, $6 > 4$) = $>$ | 6. Row A = 4, Row B = 4, more, fewer, or the same = = |
| 3. stars = 5, moons = 5, more, fewer, or the same = = | 7. Array A = 9, Group B = 7, which has more (A, $9 > 7$) = $>$ |
| 4. Row A = 7, Row B = 6, which row has more (A, $7 > 6$) = $>$, — | 8. Row A = 6, Row B = 8, Row C = 6, the two matching rows (A and C, $6 = 6$) = =, — |
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What is verified

6 of 8 problems fully machine-checked against SymPy.

— marks an answer only you can judge — problems 4, 8. The worked solution states what a correct response must contain.

Curriculum

Level: Kindergarten–Grade 1

K.CC.C.6 – *problems 1, 2, 3, 4, 5, 6, 7, 8*

Difficulty 1–4 of 5 across 27 tagged checks.

- 1.** Cats and bowls: count each, then more cats or more bowls?

Solution: Count the cats:

Count the bowls:

When counting, 3 comes before 5, so there are more **bowls**.

3
5
$3 < 5$

- 2.** Row A (6 dots) and Row B (4 dots): which has fewer?

Solution: Row A:

Row B:

4 is the smaller count, so Row **B** has fewer.

6
4
$6 > 4$

- 3.** Stars and moons: more, fewer, or the same?

Solution: Stars:

Moons:

The counts match, so the groups are the **same**.

5
5
$5 = 5$

4. Draw pairing lines between Row A (7) and Row B (6), then circle the row with more.

Solution: Row A:

7

Row B:

6

After pairing, Row A has a dot with no partner, so Row **A** has more.

7 > 6

Pairing and circling (instructor-judged): each Row A dot connects to one Row B dot, no dot used twice; one Row A dot is left unpaired, and the circle goes around Row A. The leftover dot *is* the proof of more.

5. Tam has 8 grapes, Ana has 6: who has fewer?

Solution: Tam:

8

Ana:

6

6 is the smaller count, so **Ana** has fewer grapes.

8 > 6

6. Squeezed row and spread-out row, 4 dots each: more, fewer, or the same?

Solution: Row A:

4

Row B:

4

Spreading dots out changes how the row *looks*, never how many there are. The counts match: the **same**.

4 = 4

7. Array A (3 rows of 3) and Group B (7 dots): which has more?

Solution: Array A, row by row: $3 + 3 + 3$ gives

9

Group B:

7

9 comes after 7 in the count, so Array **A** has more.

9 > 7

8. Challenge: rows of 6, 8, and 6 — find the two with the same number and tell how you know.

Solution: Row A:

6

Row B:

8

Row C:

6

Rows **A** and **C** both count to six:

6 = 6

Tell how you know (instructor-judged): accept “both counts ended at six” or “the rows pair up dot for dot with none left over.” Either the equal-counts reason or the pairing reason earns credit.